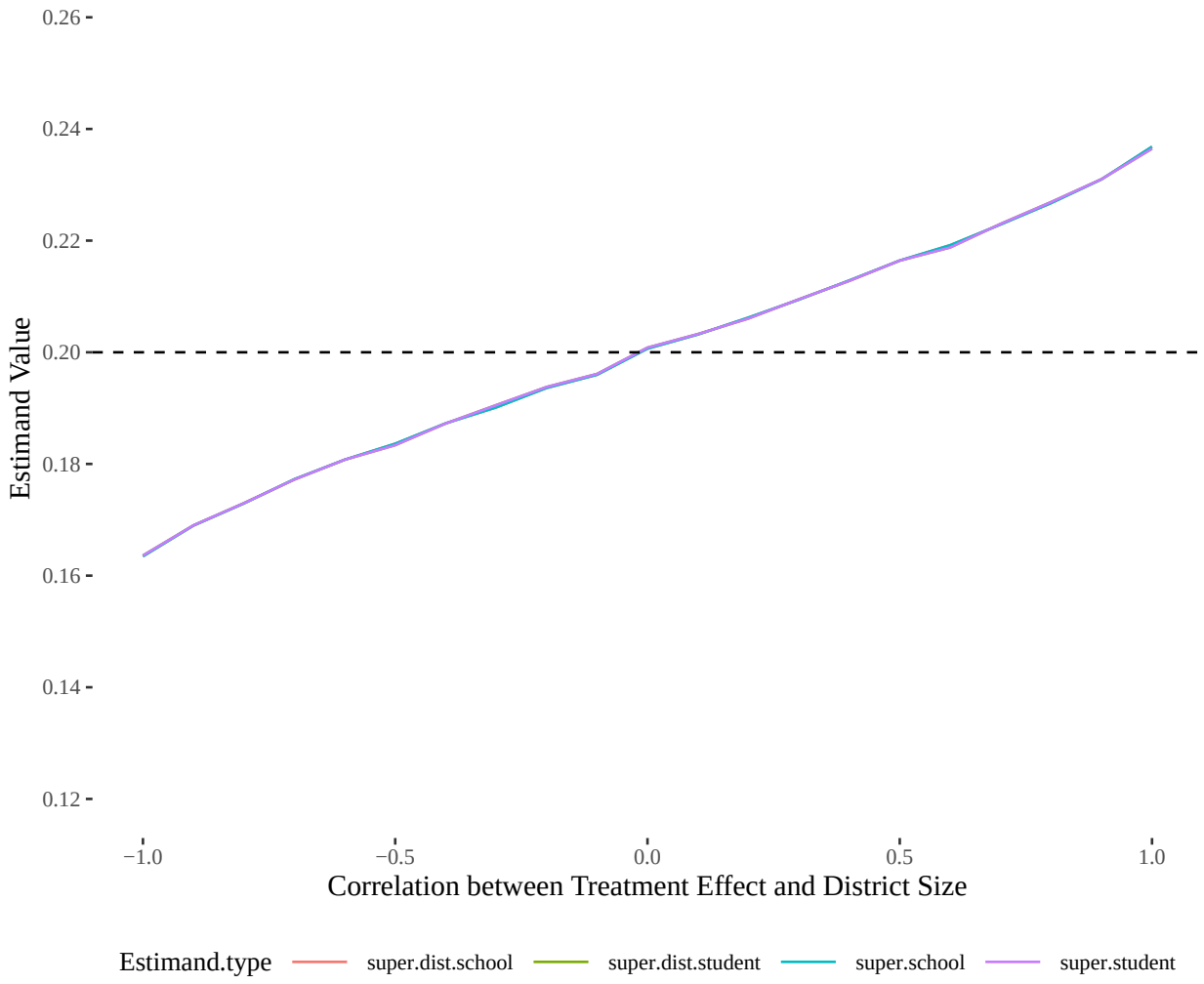
Simulation Analysis

Sophie Litschwartz

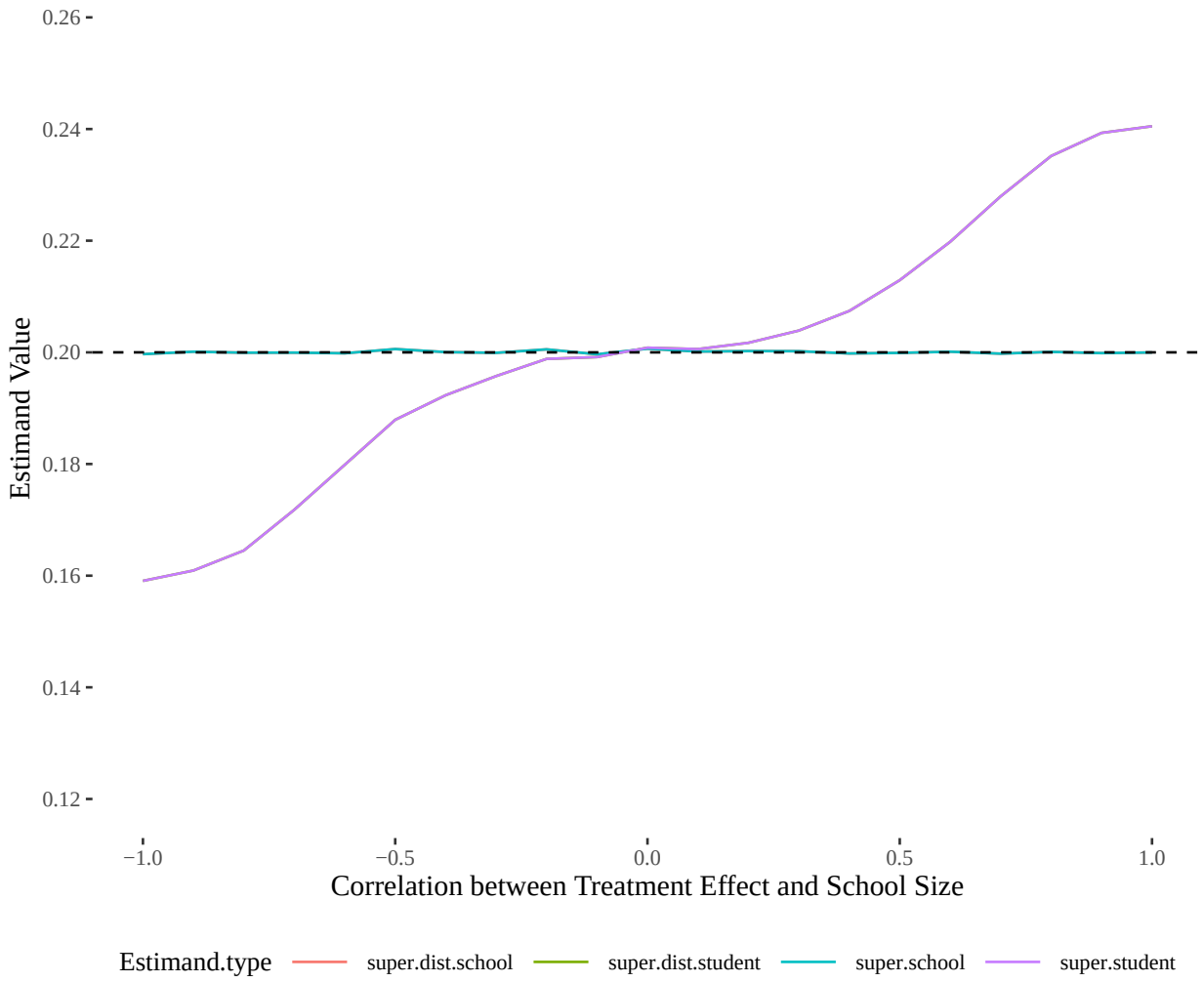
2024-08-09

When are Each of the Estimands Different From Each Other

When there is a correlation between number of schools per district and treatment effects, then the district weighted estimands are different from the school weighted estimands.



When there is a correlation between treatment effect and school size, the student weighted estimands are different from the school weighted estimands.

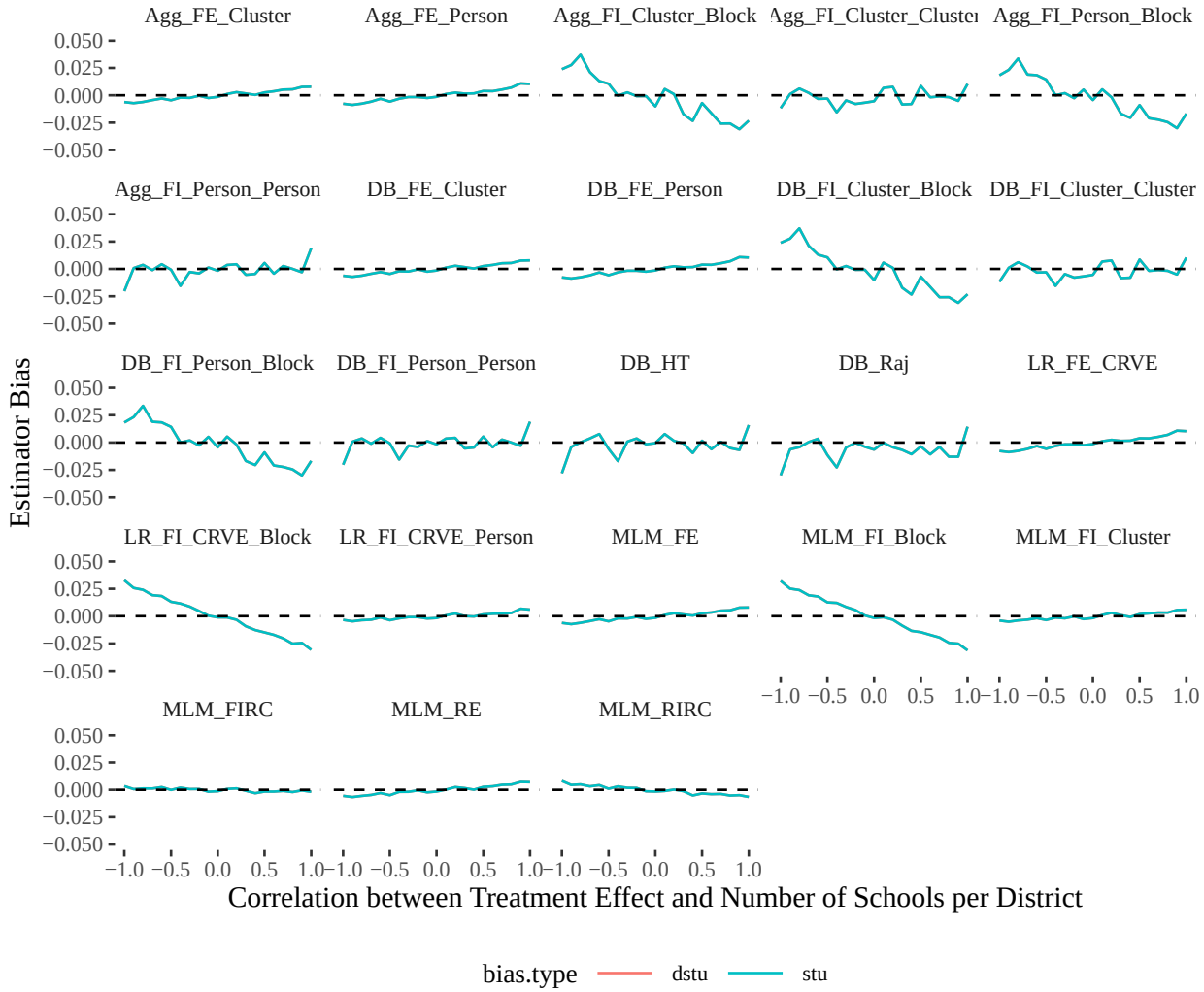


Using Average Estimator Bias for Each Estimand to Map Estimators to Estimands

For each graph all the non-varying values are set at base value. The average bias is the mean bias across 400 simulations.

The district targeted estimators are: Agg_FI_Cluster_Bloc, Agg_FI_Person_Block, DB_FI_Cluster_Block, DB_FI_Person_Block, LR_FI_CRVE_Block, and MLM_FI_Block

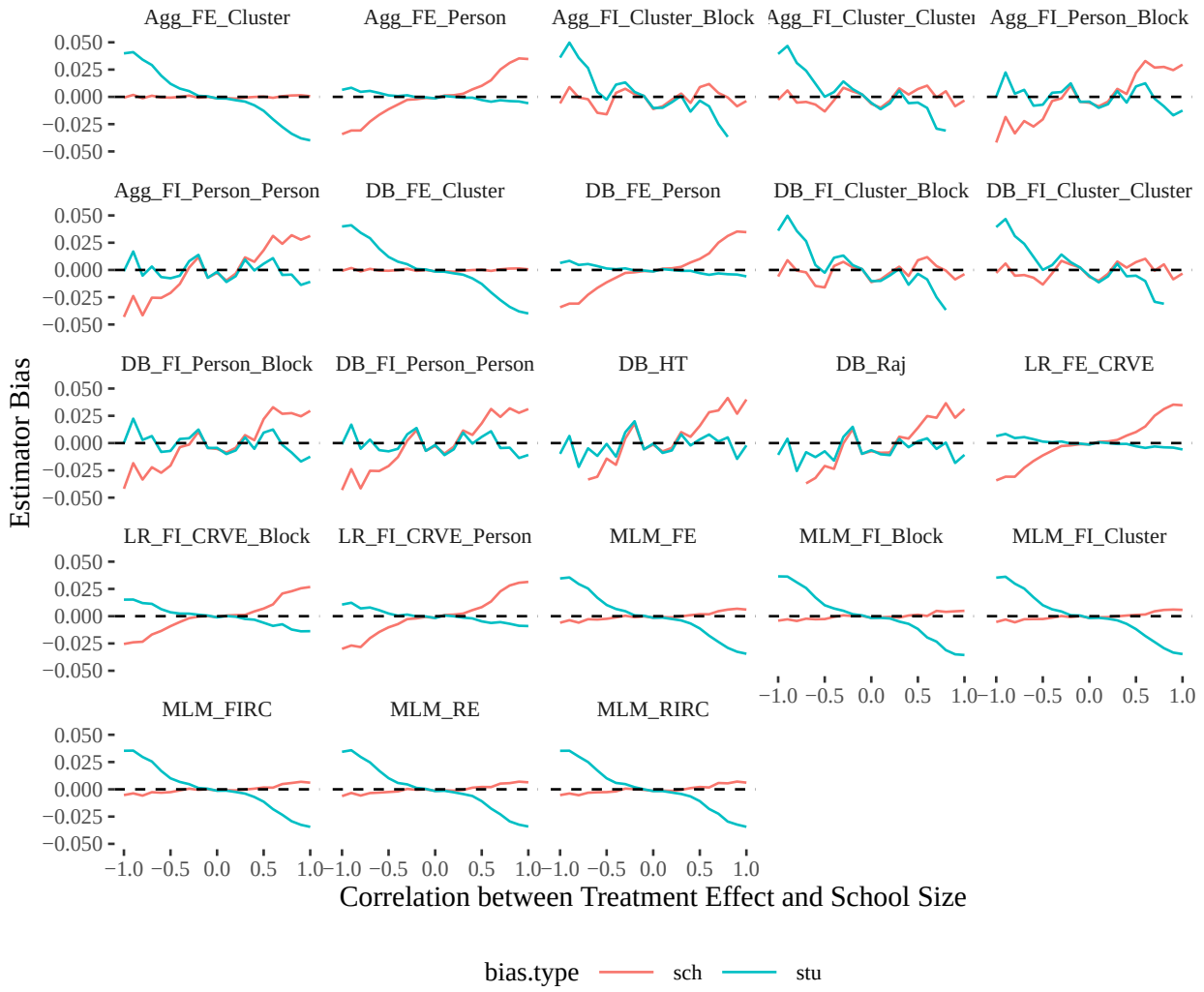
The non-district targeted estimators are: Agg_FE_Cluster, Agg_FE_Person, Agg_FI_Cluster_Cluster, Agg_FI_Person_Person, DB_FE_Cluster, DB_FE_Person, DB_FI_Cluster_Cluster, DB_FI_Person_Person, DB_HT, DB_Raj, LR_FE_CRVE, LR_FI_CRVE_Cluster, LR_FI_CRVE_Person, MLM_FE, MLM_FI_Cluster, MLM_FI_Person, MLM_FIRC, MLM_RE, MLM_RIRC



The school targeted estimators are: Agg_FE_Cluster, Agg_FI_Cluster_Block, Agg_FI_Cluster_Cluster, DB_FE_Cluster, DB_FI_Cluster_Block, DB_FI_Cluster_Cluster, MLM_FE, MLM_FI_Block, MLM_FI_Cluster, MLM_FI_Person, MLM_FIRC, MLM_RE, MLM_RIRC

The student targeted estimators are: Agg_FE_Person, Agg_FI_Person_Person, Agg_FI_Person_Block,

DB_FE_Person, DB_FI_Person_Block, DB_FI_Person_Person, DB_HT, DB_Raj, LR_FE_CRVE, LR_FI_CRVE_Block, LR_FI_CRVE_Cluster, LR_FI_CRVE_Person

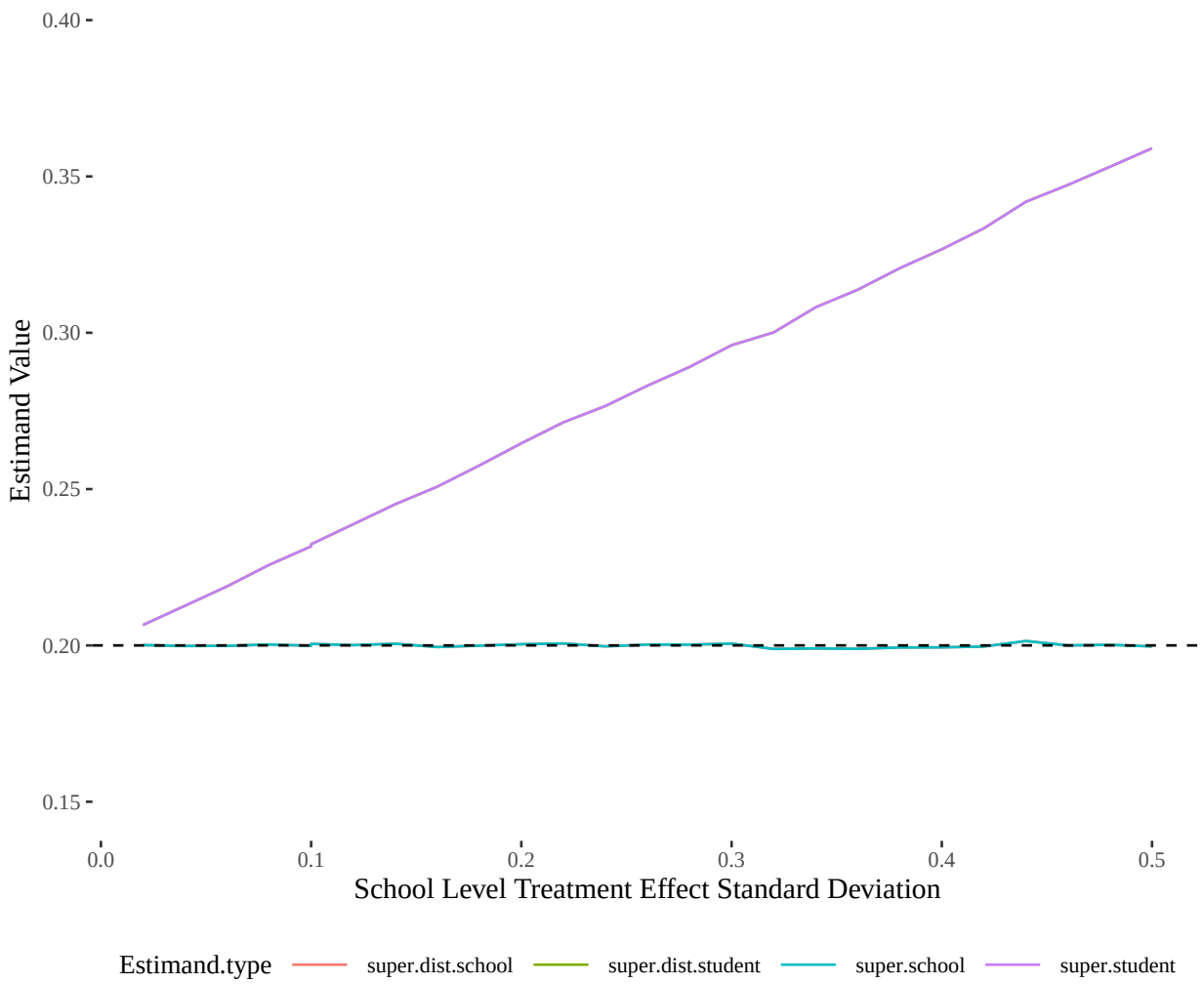


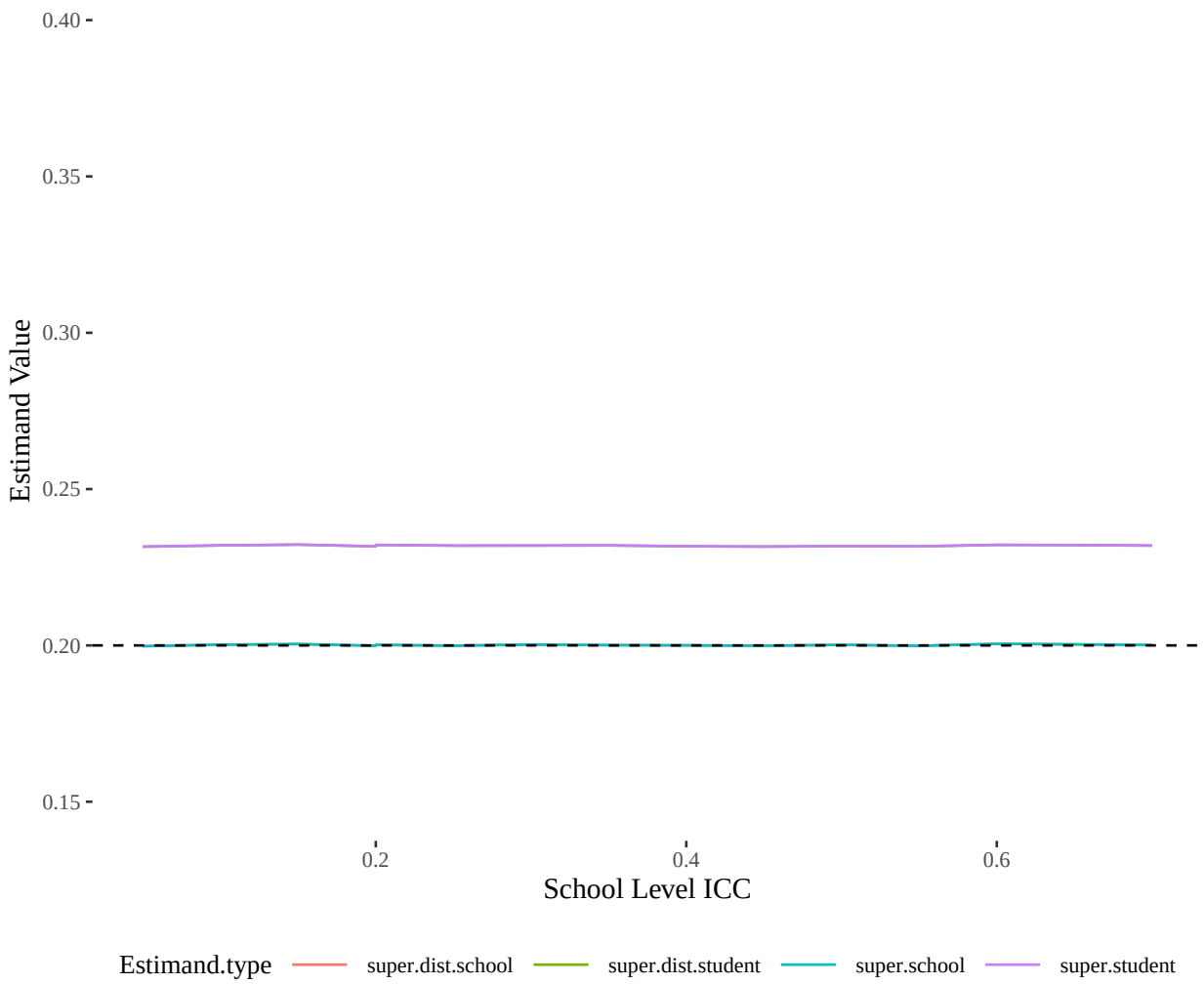
Mapping Estimators to Estimands

These simulations have the same base parameter values as the others but the correlation between school size and treatment effect is set at .75, with the idea that this should induce enough of a difference between the school and students weighted estimands.

Re-Looking at When Are The Estimands Different

Conditional on a correlation between the treatment effect and school size. The only thing that impacts the delta between the school and student weighted estimands is the school level treatment effect standard deviation.


0.40
0.35
0.30
0.25
0.20
0.15
Estimand Value
0.0
0.1
0.2
0.3
0.4
0.5
School Level Treatment Effect Standard Deviation
Estimand.type
super.dist.school
super.dist.student
super.school
super.student



Which Estimators Produce the Same Values or Close to the Same Values

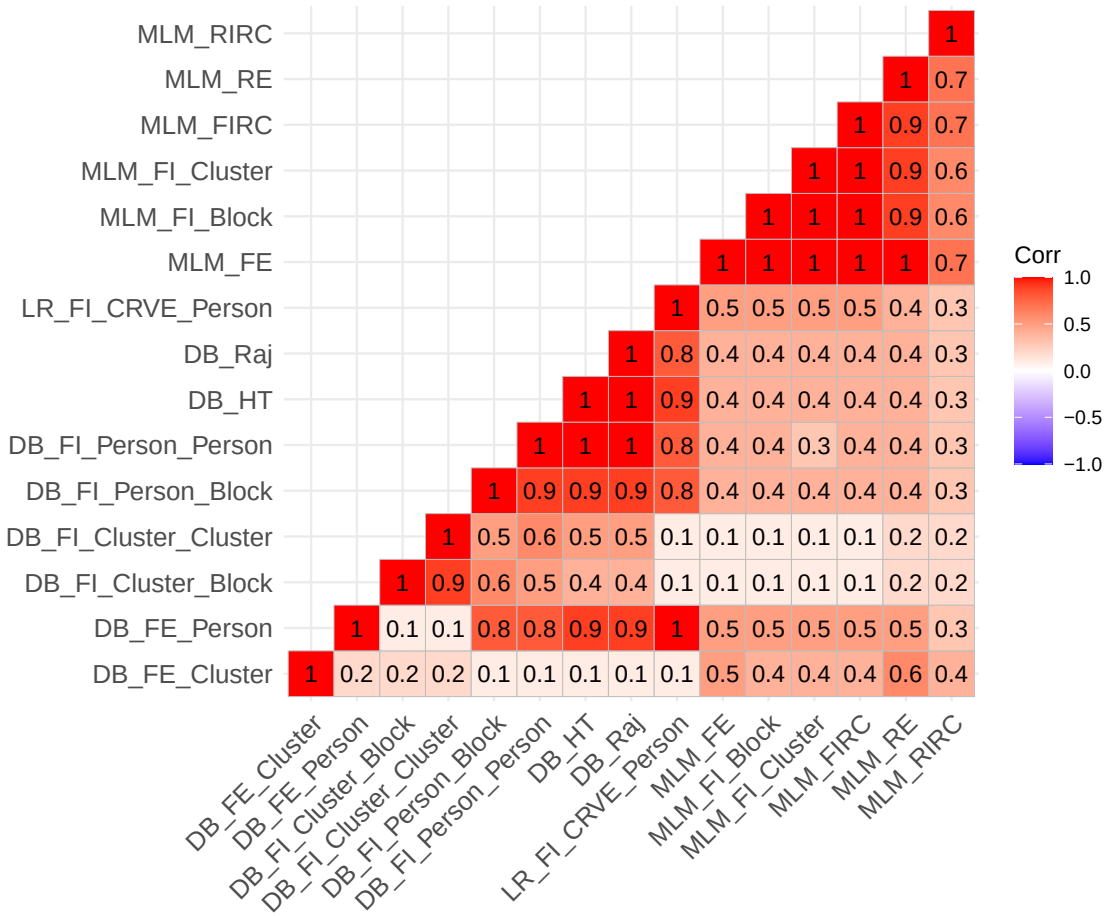
In all simulations that follow I've also increased the school and district treatment effect standard deviations to .3 in order to maxmize the differences between the school and student weighted estimands. Each row in the table are estimators that produce the same values

```
## Warning: Using an external vector in selections was deprecated in tidyselect 1.1.0.
## i Please use `all_of()` or `any_of()` instead.
##   # Was:
##   data %>% select(var)
##
##   # Now:
##   data %>% select(all_of(var))
##
## See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

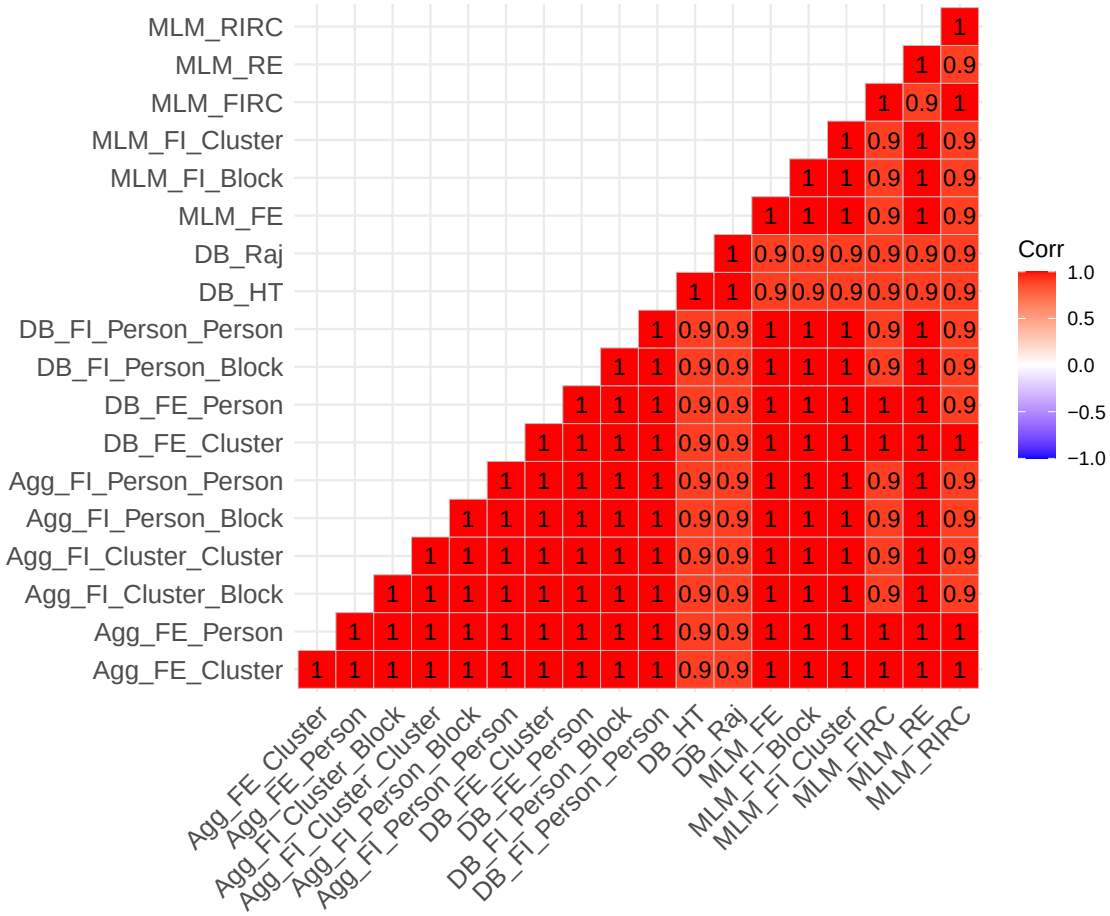
grp	.
1	Agg_FE_Cluster, DB_FE_Cluster
2	Agg_FE_Person, DB_FE_Person, LR_FE_CRVE
3	Agg_FI_Cluster_Block, DB_FI_Cluster_Block
4	Agg_FI_Cluster_Cluster, DB_FI_Cluster_Cluster
5	Agg_FI_Person_Block, DB_FI_Person_Block
6	Agg_FI_Person_Person, DB_FI_Person_Person
7	DB_HT
8	DB_Raj
9	LR_FI_CRVE_Block
10	LR_FI_CRVE_Person
11	MLM_FE
12	MLM_FI_Block
13	MLM_FI_Cluster
14	MLM_FIRC
15	MLM_RE
16	MLM_RIRC

grp	.
1	Agg_FE_Cluster
2	Agg_FE_Person, LR_FE_CRVE
3	Agg_FI_Cluster_Block, DB_FI_Cluster_Block
4	Agg_FI_Cluster_Cluster, DB_FI_Cluster_Cluster
5	Agg_FI_Person_Block
6	Agg_FI_Person_Person
7	DB_FE_Cluster
8	DB_FE_Person
9	DB_FI_Person_Block
10	DB_FI_Person_Person
11	DB_HT
12	DB_Raj
13	LR_FI_CRVE_Block
14	LR_FI_CRVE_Person
15	MLM_FE
16	MLM_FI_Block
17	MLM_FI_Cluster
18	MLM_FIRC
19	MLM_RE
20	MLM_RIRC

This is the correlation across estimators once the ones that produce the same results are dropped



This is the correlation across standard error estimates once the ones that produce the same results are dropped



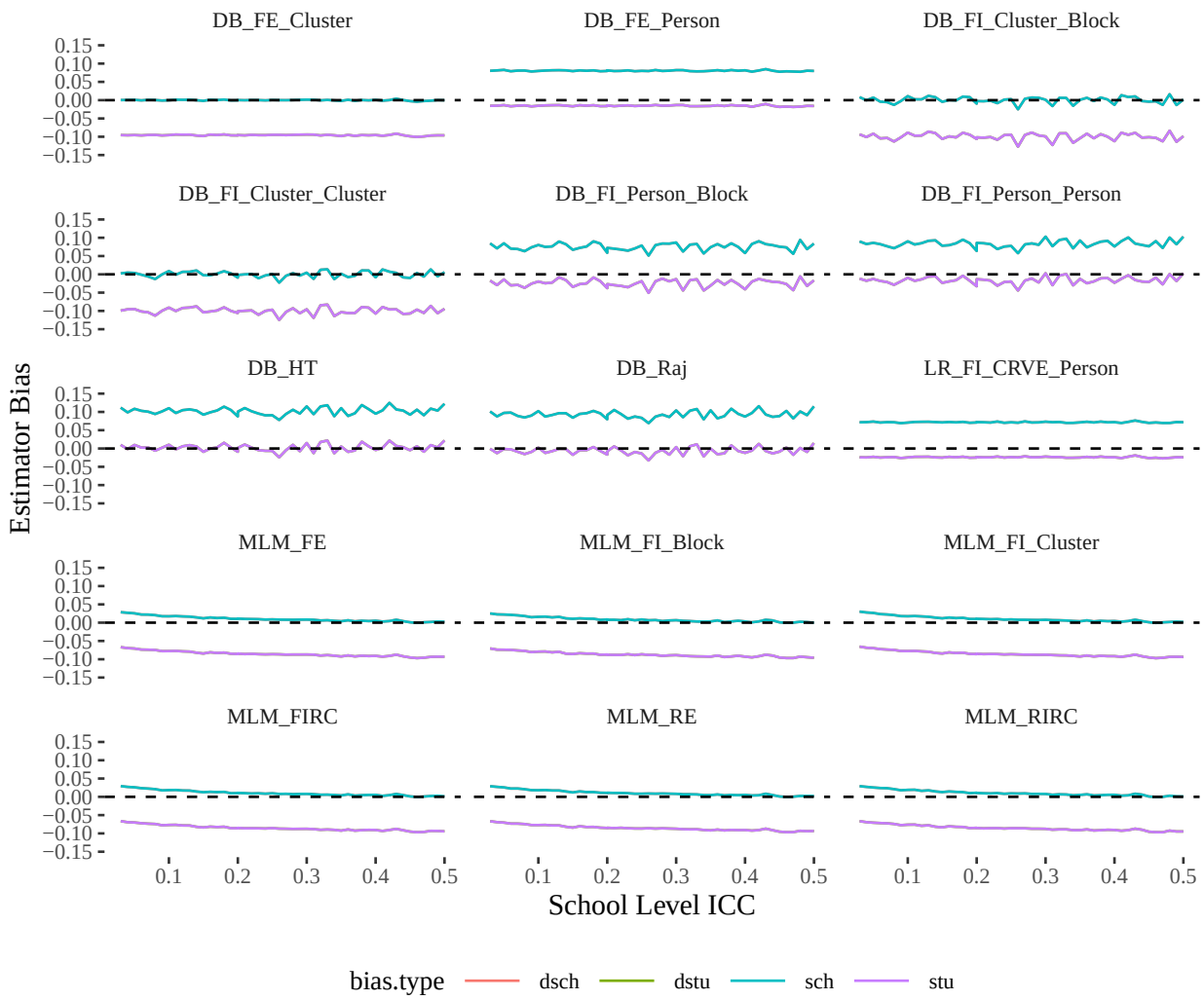
Confirming School versus Student targeted estimators

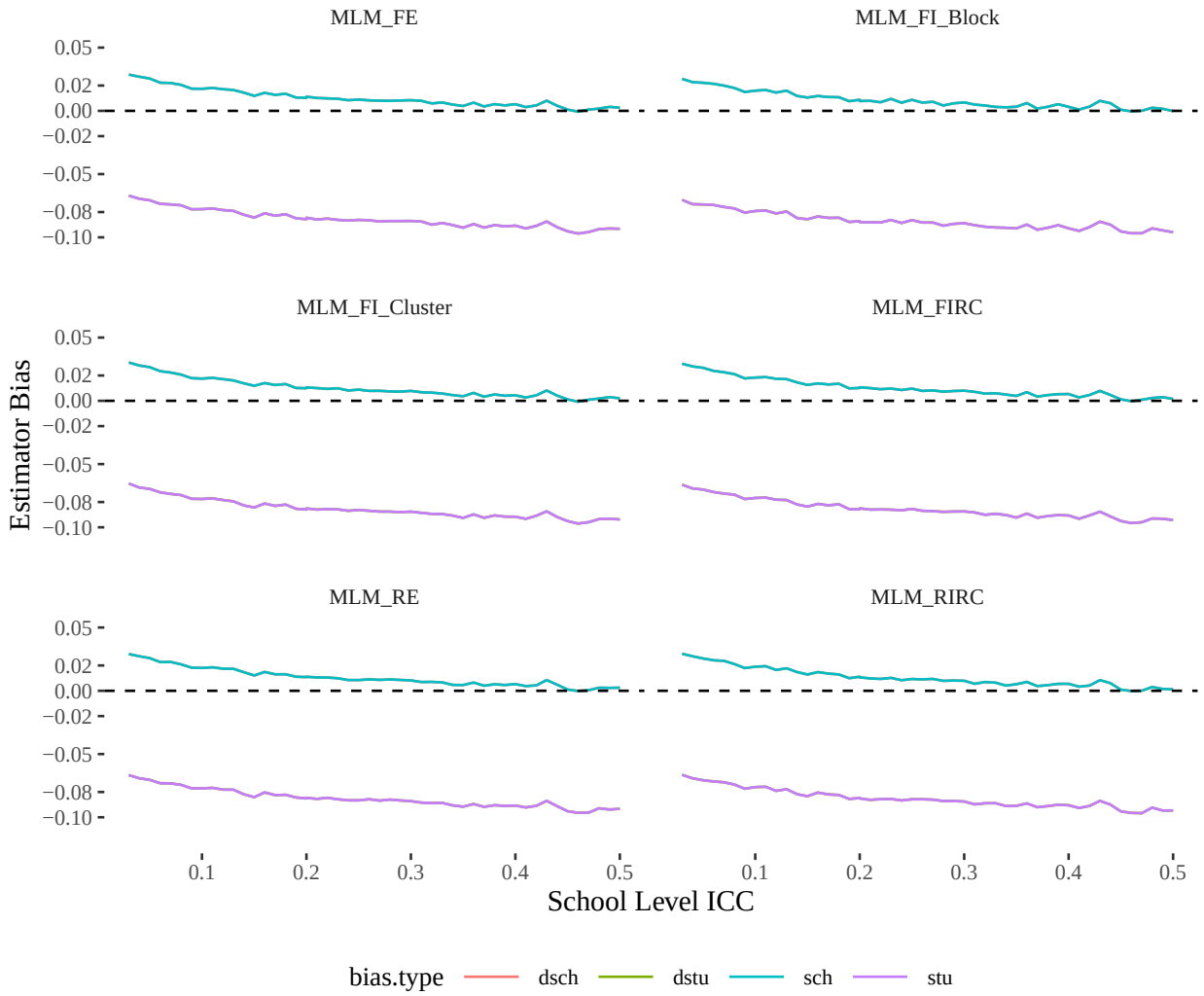
The same as in the analysis above, consistently:

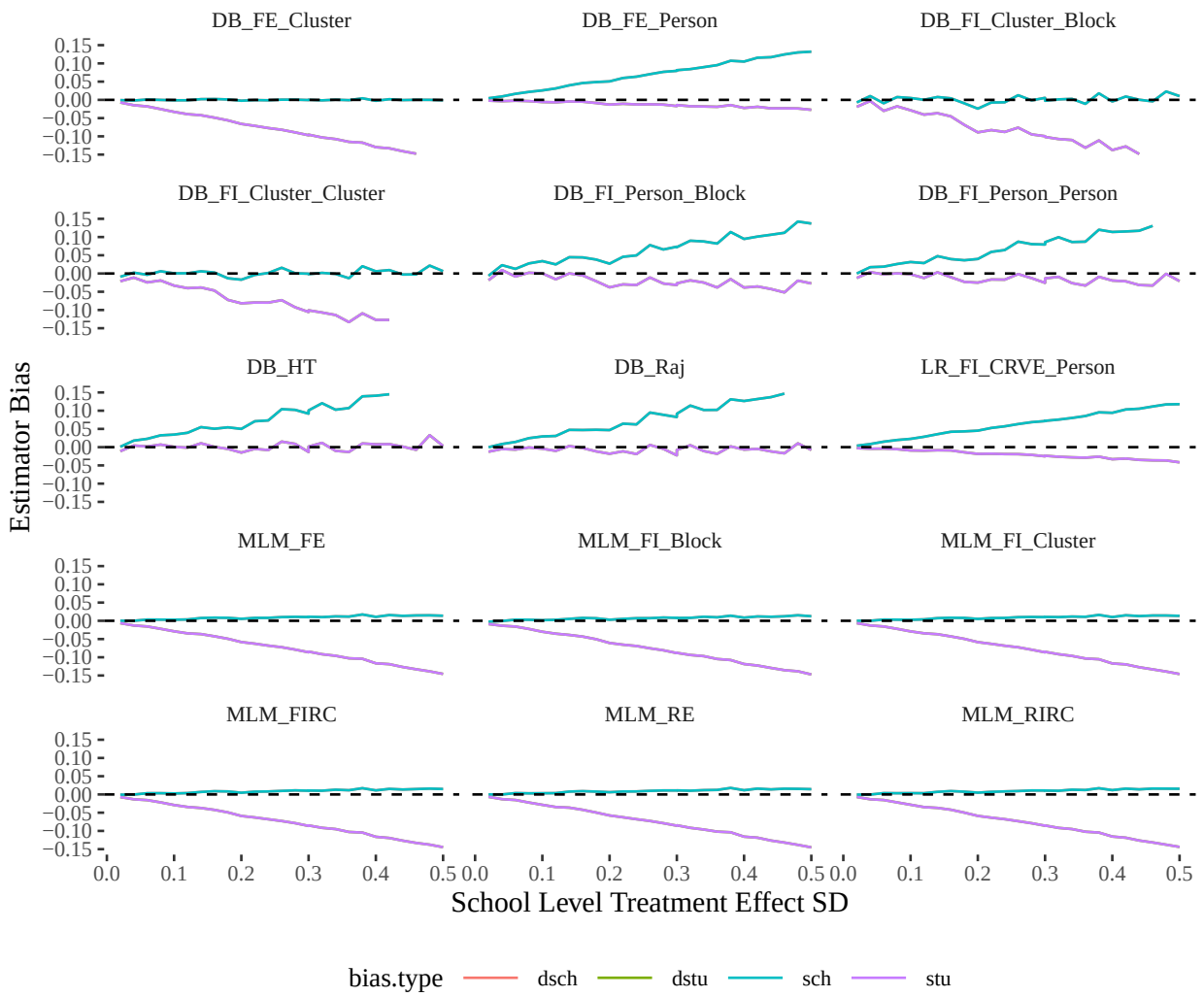
The school targeted estimators are: DB_FE_Cluster, MLM_FE, MLM_FI_Block, MLM_FI_Cluster, MLM_FI_Person, MLM_FIRC, MLM_RE, MLM_RIRC

The student targeted estimators are: DB_FE_Person, DB_FI_Person_Block, DB_FI_Person_Person, DB_HT, DB_Raj,LR_FI_CRVE_Cluster, LR_FI_CRVE_Person

*The MLM estimators aren't really student targeted when the ICC is low, they're really closest to the school weighted estimand the whole time. However, they do seem to be weighting between the student and school estimands and as the school ICC goes up they become more school weighted.


DB_FE_Cluster
DB_FE_Person
DB_FI_Cluster_Block
DB_FI_Cluster_Cluster
DB_FI_Person_Block
DB_FI_Person_Person
DB_HT
DB_Raj
LR_FI_CRVE_Person
MLM_FE
MLM_FI_Block
MLM_FI_Cluster
MLM_FIRC
MLM_RE
MLM_RIRC
Estimator Bias
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0.1
0.2
0.3
0.4
0.5
School Level ICC
bias.type
dsch
dstu
sch
stu


MLM_FE
MLM_FI_Block
MLM_FI_Cluster
MLM_FIRC
MLM_RE
MLM_RIRC
Estimator Bias
0.05
0.02
0.00
−0.02
−0.05
−0.08
−0.10
0.1
0.2
0.3
0.4
0.5
School Level ICC
bias.type
dsch
dstu
sch
stu

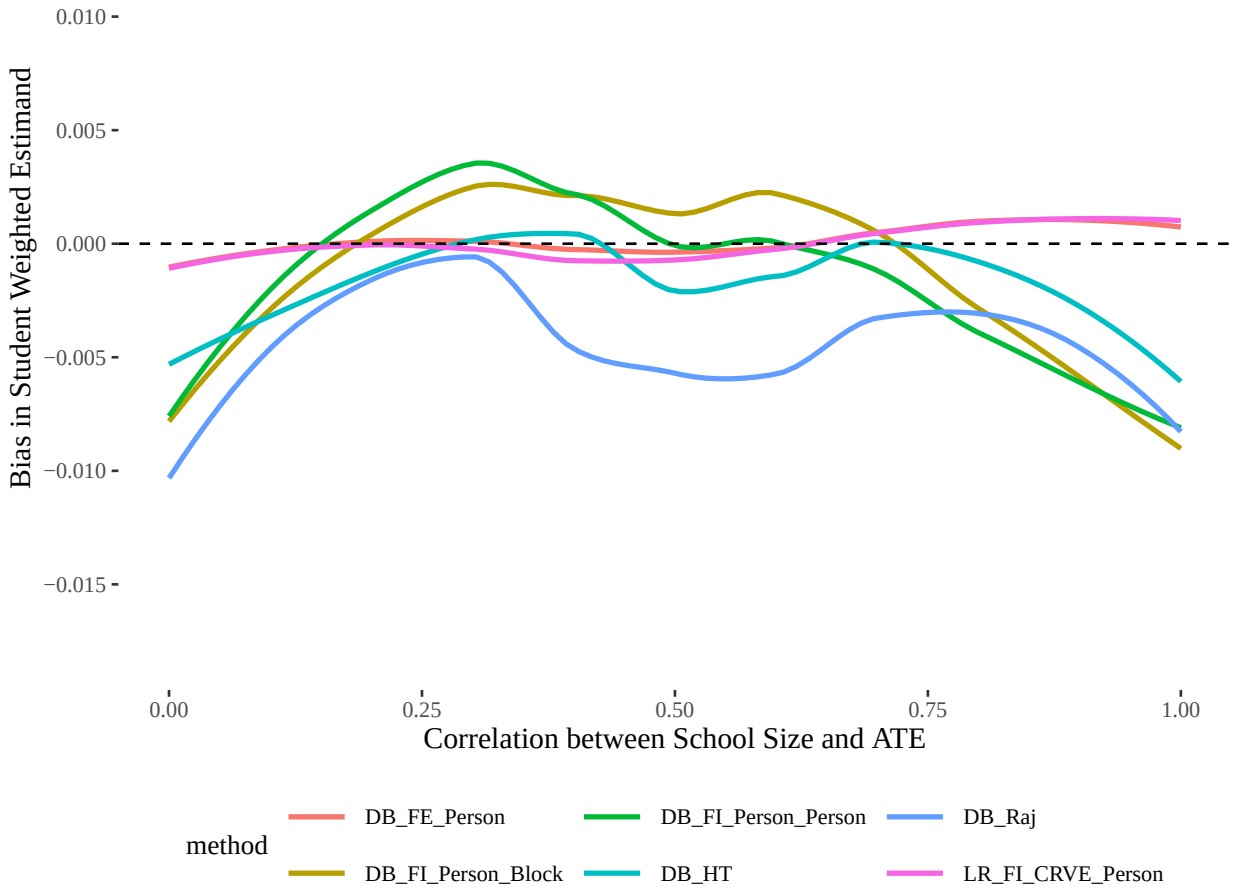


How biased are the student weighted estimators as the correlation between school size and treatment effect increases

Here are simulation with the non-varying parameter values back at the base values. Even large correlations between the ATE and the school size do not lead to substantially biased estimates from the student targeted estimators.

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
## Warning: Removed 6 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```



Simulation Parameters

The simulation currently takes in the following parameters. The base value is the value across simulations where the value doesn't vary.

K: The total number of districts
K base value: 10

J.bar: The average number of schools per district. This must be at least 6.
J.bar base value: 10

p.bar: The average proportion of schools treated in a district
p.bar base value: 0.5

ATE.district: The average treatment effect
ATE.district base value:

sd.district: The district standard deviation in the treatment effect
sd.district base value: 0.1

ICC.district: The district level ICC
ICC.district base value: 0.2

variable.J: Indicator for whether the number of schools per district varies
variable.J base value: TRUE

variable.p: Indicator for whether the proportion of schools treated per district varies
variable.p base value: FALSE

n.bar: The average number of students per school.
n.bar base value: 25

ICC.school: The school level ICC
ICC.school base value: 0.2

variable.n: Indicator for whether the number of students per school varies
variable.n base value: TRUE

size.impact.cor.district: Correlation between the number of schools in a district and treatment effect
size.impact.cor.district base value: 0

proptx.impact.cor.district: Correlation between the proportion of schools treated in a district and treatment effect
proptx.impact.cor.district base value: 0

size.intercept.cor.district:Correlation between the number of schools in a district and district average outcome value
size.intercept.cor.district base value: 0

size.impact.cor.school: Correlation between the number of students per schools and treatment effect
size.impact.cor.school base value: 0